

Earning clinical trust before scaling autonomy

A working prototype and AI transformation thesis for clinical workflows

OBSERVE

01

ASSIST

02

EARN TRUST

03

BUSINESS PERSPECTIVE

The bottleneck is not intelligence. It is trusted workflow adoption.

01 Today: fragmented work

Repeated symptom collection
Context rebuilt at review time
Manual test selection and documentation
Generic AI that does not reflect how a doctor works

02 Constraint: clinical trust

A useful assistant must know when to stop
Doctors need visibility and correction rights
Safety cannot depend on one model response
Practice differs by doctor and specialty

03 Opportunity: governed learning

Structure cases before review
Learn from accepted and rejected decisions
Turn reviewed behavior into reusable capability
Scale one platform without flattening practice

Business hypothesis

Reduce repetitive work while increasing the evidence available to govern each next level of assistance.

No ROI is assumed: Shadow mode establishes the operational baseline before value is claimed.

RESPONSIBLE AI CONTEXT

Singapore guidance points to a lifecycle, not a launch checklist.

- 1 Human agency** Appropriate human involvement and clear accountability
- 2 Clinical safety** Risk-based controls, validation and escalation
- 3 Transparency** Explainable decisions and understandable communication
- 4 Data governance** Purpose, quality, access, privacy and traceability
- 5 Lifecycle control** Monitoring, robustness, change management and review

SOURCE HIERARCHY

Healthcare-specific

AIHGle (2021) as historical sector context; current HSA software medical-device lifecycle and SaMD-CDSS classification guidance where applicable.

Cross-sector governance

PDPC Model AI Governance Framework: human-centric; explainable, transparent and fair.

DESIGN ALIGNMENT, NOT COMPLIANCE

GOVERNANCE TRACEABILITY

The prototype turns principles into controls - and exposes what is still missing.

Principle	Product control	Prototype evidence	Production gap
Human oversight	Approve / modify; bounded modes	Doctor review + audit events	RBAC and governance board
Clinical safety	Deterministic override; fail-safe	RED forced; failures become AMBER	Clinical validation + hazard analysis
Data governance	Candidate approval; approved-only use	Memory status and provenance	Consent, retention, encryption
Monitoring	Evaluation suite; audit trail	32 synthetic cases + metrics	Drift, incidents, external validation

IMPLEMENTED

SYNTHETICALLY TESTED

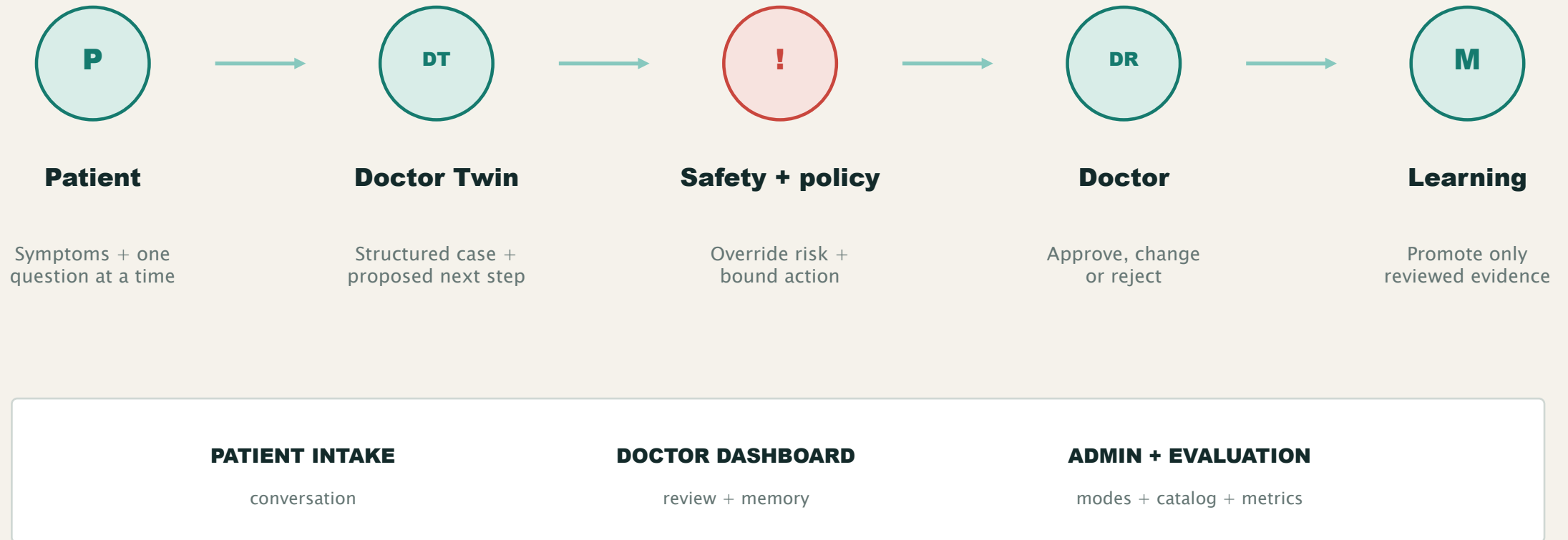
PRODUCTION WORK

Every claim uses one of these labels.

Architecture alignment reduces risk; it does not replace clinical, privacy, security or regulatory review.

WORKING PRODUCT

One workflow connects patient context, doctor judgment and governed learning.



Implemented with synthetic patients and doctors; real clinical integrations are mocked.

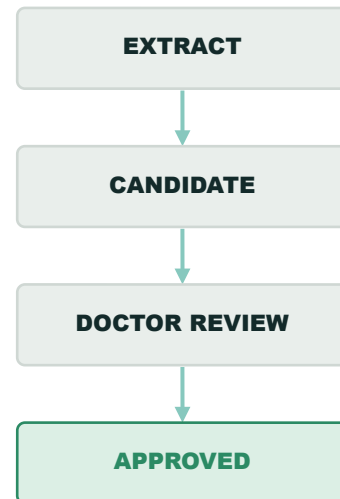
DOCTOR-SPECIFIC LEARNING

The system learns two different things - and governs them differently.

1 EPISODIC MEMORY

What this doctor prefers

Communication style
Escalation rules
Clinical patterns
Workflow preferences



2 TEST-DECISION EVIDENCE

How this doctor acts in context

SHADOW: selected

COPILOT: rejected

COPILOT: accepted

COPILOT: corrected

Only finalized Shadow/Copilot decisions train future recommendations. Autonomous decisions never train themselves.

Doctor benefit: learned assumptions remain visible and correctable.

DATA MANAGEMENT

Shared foundation. Patient-bound context. Doctor-private learning.

SHARED ACROSS DOCTORS**Organization-governed reference layer**

Diagnostic-test catalog
Safety rules and escalation logic
Deployment-policy definitions
Clinical output schemas and base prompts
Evaluation cases and aggregate quality metrics

CONSISTENT CONTROL PLANE**PATIENT-SPECIFIC****Available only for an authorized care relationship**

Patient profile, conditions, medications and allergies
Encounter conversation and assessment
Prior visit summaries
Appointments and encounter-linked audit history

CARE-CONTEXT BOUNDARY**DOCTOR-SPECIFIC****Never pooled across doctors in the prototype**

Doctor profile and default mode
Approved memories and preferences
Shadow selections and Copilot labels
Doctor-specific test evidence and acceptance history
Doctor case queue and scheduling

ISOLATED BY doctor_id

Prototype limitation: authentication, RBAC, consent and tenancy are not implemented; production must enforce these boundaries, not merely model them.

PILOT ECONOMICS

Shadow mode creates the baseline; Copilot must earn the delta.

EFFICIENCY

- **Intake time**
Submission to review-ready
- **Review time**
Open to approve / modify
- **Throughput**
Cases per session

ADOPTION

- **Acceptance**
Recommendations retained
- **Override rate**
Risk / action / test edits
- **Active use**
Weekly participating doctors

SAFETY + QUALITY

- **Red-flag recall**
Urgent cases detected
- **Fallback rate**
Provider / output failures
- **Disagreement**
Clinically significant variance

BUSINESS

- **Unit cost**
Cost per completed intake
- **Capacity**
Review minutes released
- **Patient flow**
Wait to next action

Expand only when workflow value improves without weakening predefined safety guardrails.

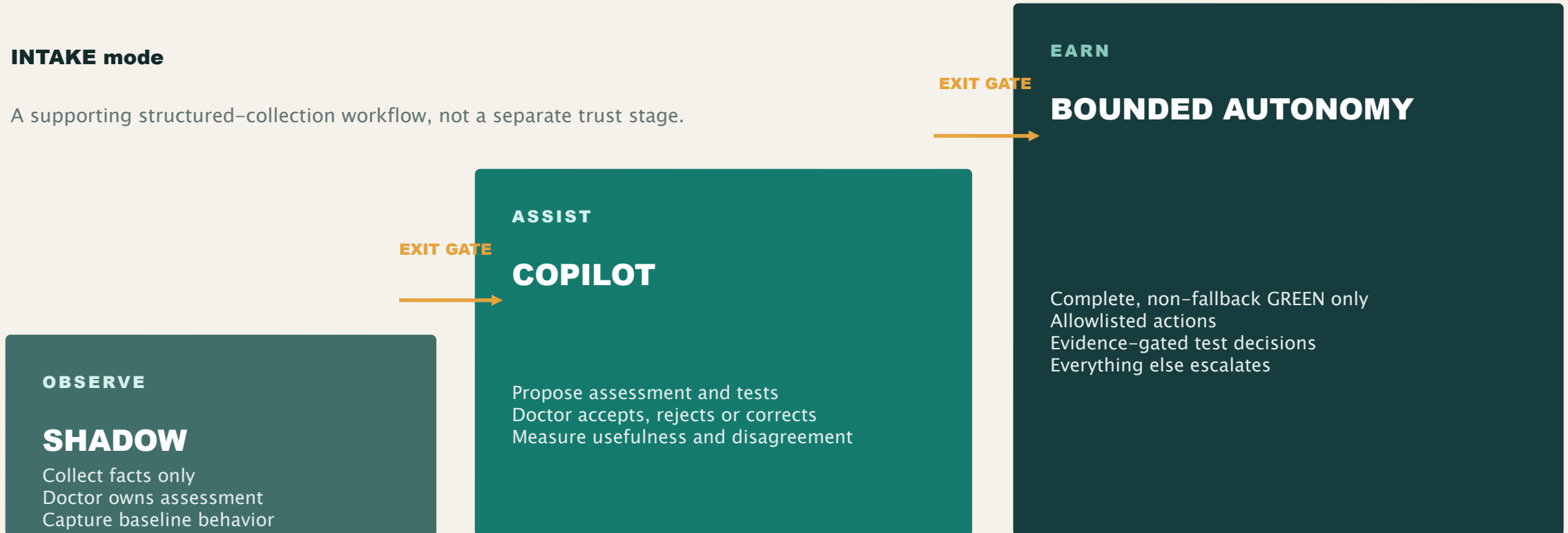
Targets should be agreed with clinical and operational owners before the pilot starts.

ADOPTION STRATEGY

Trust is earned in stages - with a measurable exit gate at each step.

INTAKE mode

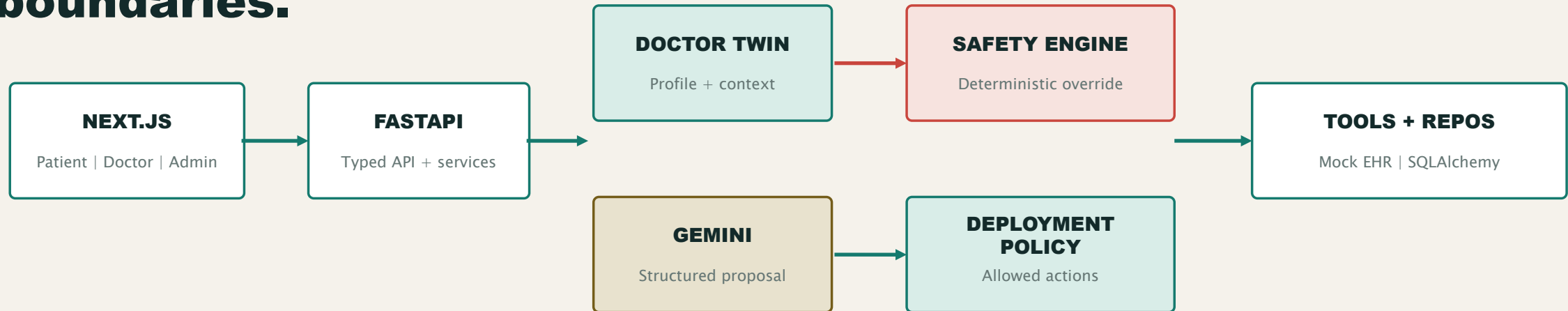
A supporting structured-collection workflow, not a separate trust stage.



Mode graduation is a clinical governance decision based on evidence - never an automatic model choice.

TECHNICAL PERSPECTIVE

Probabilistic reasoning sits inside deterministic control boundaries.



APPROVED MEMORY	DOCTOR EVIDENCE	AUDIT EVENTS	EVALUATION
Doctor-scoped	Finalized labels	Append-only trail	32 synthetic cases

ONE REUSABLE TWIN Personalized at runtime by doctor profile, approved memory and doctor-scoped evidence.

Prototype: SQLite, keyword retrieval and mocked clinic adapter. Production: identity, PostgreSQL, integrations and observability.

MODE 1 | OBSERVE

Shadow mode: assist with intake, observe decisions.

PATIENT JOURNEY

DASHBOARD OUTCOME

LEARNING
RULE

VALUE TODAY

Less clerical intake work and a review-ready case, while the doctor keeps all clinical judgment.

LEARNING
CREATED

A baseline of doctor-authored decisions. Memory candidates remain inactive until the doctor approves them.

Shadow does not train from raw model output: only reviewed doctor decisions and explicitly approved memories influence future behavior.

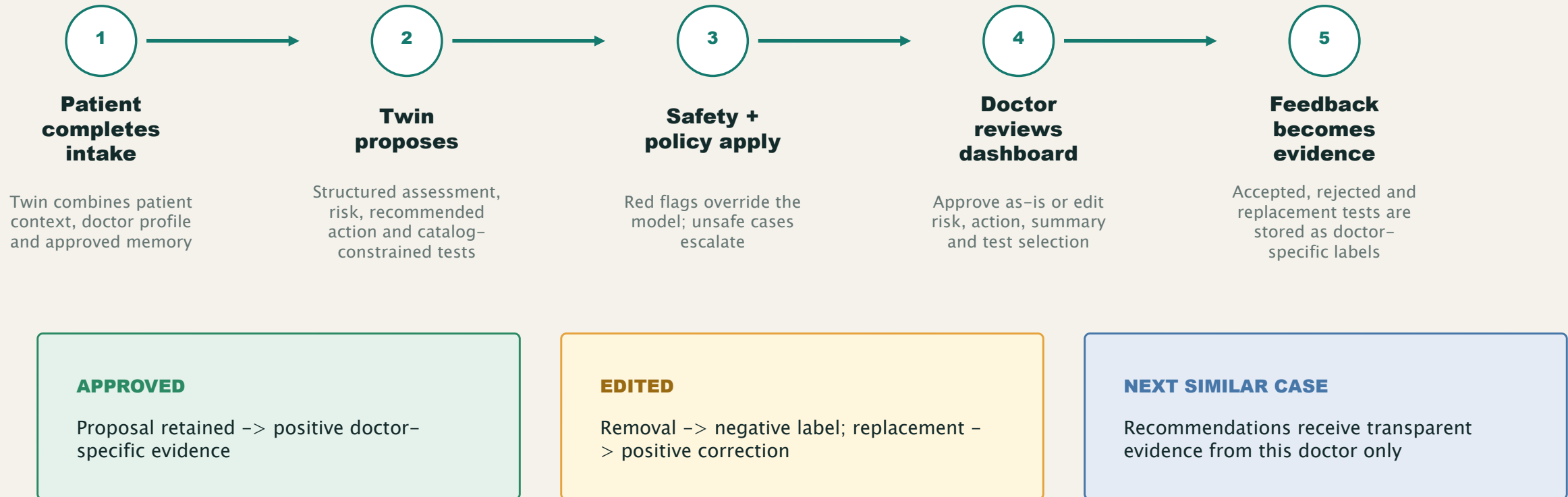
MODE 2 | ASSIST

Copilot mode: recommend, review, and learn response.

PATIENT JOURNEY

DOCTOR REVIEW

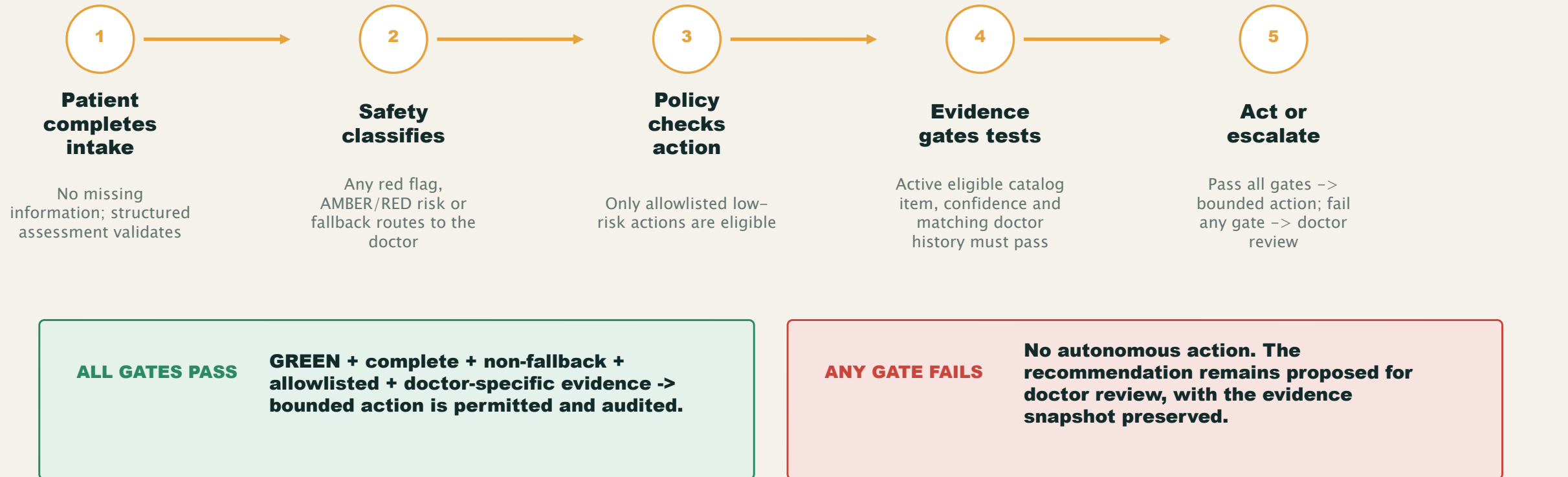
LEARNING LOOP



Copilot shortens the decision loop without removing it: every clinical proposal remains reviewable and correctable.

MODE 3 | BOUNDED AUTONOMY

Autonomous mode: act only inside a narrow envelope.

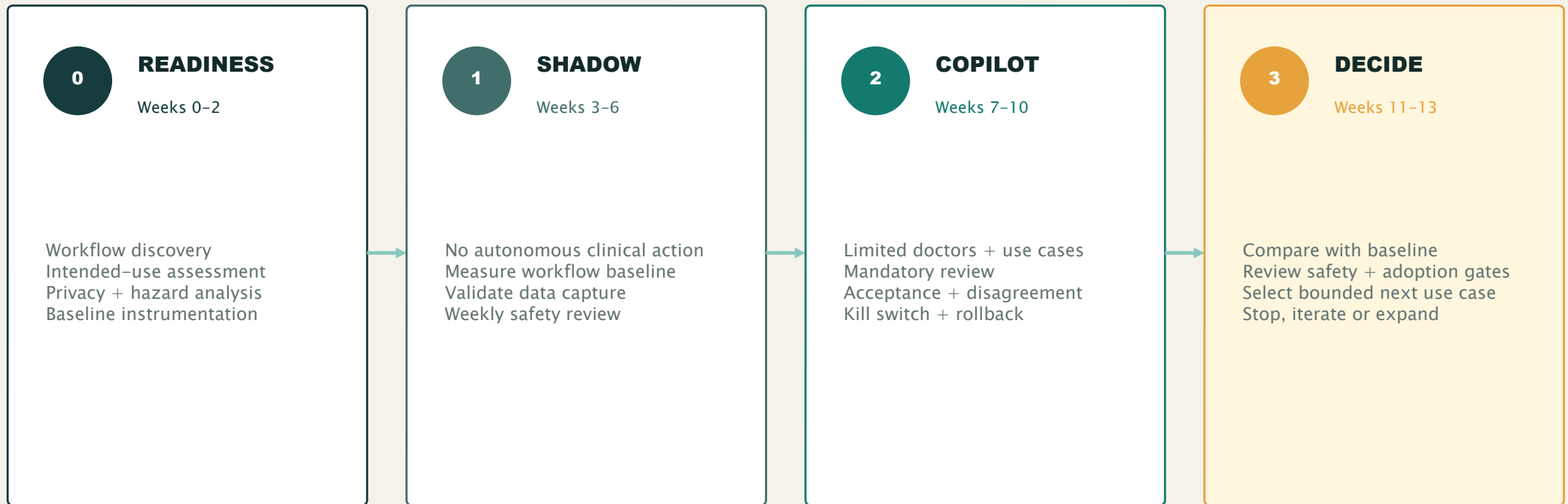


LEARNING RULE Autonomous actions do not create training evidence; future behavior changes only through reviewed Shadow/Copilot decisions.

Autonomous decisions never train themselves; only subsequent doctor-reviewed Shadow/Copilot evidence can change future recommendations.

EXECUTION ROADMAP

A 90-day pilot should buy evidence before it buys autonomy.

**Production prerequisites**

Identity + RBAC | privacy + consent | validated clinical rules | monitoring | PostgreSQL | real EHR/lab adapters

Pilot scope should be narrow enough that every disagreement can be reviewed and acted on.

THE END STATE

A governed learning layer around clinical work.

Not a digital replacement for a doctor. A system that makes reviewed clinical behavior reusable, measurable and auditable.

01 BETTER PREPARATION

More complete cases before review

02 LESS REPETITION

Administrative and cognitive work reduced

03 EVIDENCE-LED TRUST

Assistance expands only where outcomes support it

Start narrow. Measure in Shadow. Earn Copilot trust. Scale only where evidence supports it.

APPENDIX A

Singapore guidance: sources and careful applicability.

- 1 PDPC Model AI Governance Framework, 2nd edition** **21 Jan 2020**
pdpc.gov.sg/.../singapores-approach-to-ai-governance
Human-centric; explainable, transparent and fair. Sector-agnostic implementation guidance.
- 2 HSA GL-04-R4: Software Medical Devices - A Life Cycle Approach** **Dec 2025**
hsa.gov.sg/medical-devices/guidance-documents
Lifecycle guidance where intended use meets software medical-device definitions.
- 3 HSA GL-07-R2: Risk Classification SaMD-CDSS** **Jul 2025**
hsa.gov.sg/medical-devices/guidance-documents
Risk classification where intended use meets SaMD/CDSS definitions.
- 4 Artificial Intelligence in Healthcare Guidelines (AIHGle)** **2021 context**
[Authoritative archive verification pending](#)
Historical healthcare-specific reference. Verify authoritative archived edition before external circulation.

APPENDIX B

Safety and governance controls implemented in the prototype.

Deterministic red flags

Patient-authored statements are checked independently; red flags force RED and escalation.

Fail-safe handling

Provider or validation failure becomes a persisted AMBER case for review.

Mode permissions

Each deployment mode receives explicit allowed actions.

Human authority

Doctors approve or modify assessment, tests and closure; actions are audited.

Approved-only memory

Extracted memories remain candidates until explicit approval.

Evidence-gated tests

Autonomy requires complete GREEN intake, eligibility, confidence and evidence.

No self-training

Autonomous decisions are excluded from training evidence.

Synthetic evaluation

32 cases measure red flags, escalation, output validity and question coverage.

Software controls demonstrated with synthetic data are not evidence of clinical validation.

APPENDIX C

Current capability versus production end state.

WORKING NOW

Patient / doctor / admin UI
Structured intake
Doctor review and closure
Deterministic override
Governed memory
Doctor-specific test evidence
Synthetic evaluation

PROTOTYPE LIMIT

Synthetic data only
SQLite persistence
Keyword retrieval
Mock EHR adapter
Heuristic thresholds
No real laboratory
No clinical validation

PRODUCTION WORK

Authentication and RBAC
Consent and retention
Encryption and keys
PostgreSQL and migrations
Observability
Validated clinical rules
Real EHR / lab integration

Priority: identity/privacy and clinical safety validation before broader autonomous behavior.